

Sri Lanka Institute of Information Technology



SE2030: Software Engineering
Year 2, Semester 2, 2026

PROJECT PROPOSAL REPORT

Web-Based Vehicle Rental System

Lanka Ride Rentals

Group ID: 2026-Y2-S1-MLB-B13G2-05

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Submission Date: 06th of August 2026

1. Introduction

1.1 Project Overview

Lanka Ride Rentals is a vehicle rental firm with three branches, in Colombo, Kandy and Galle. The company currently handles vehicle availability, bookings and maintenance manually, using spreadsheets, phone calls between locations and paper-based service records. This results in double booking when the same vehicle is booked at different branches, lack of visibility of the current location of a vehicle, deferred updates when a vehicle is under maintenance and time – consuming manual production of reports for the management.

To replace these manual processes the Web-Based Vehicle Rental Service is proposed to provide a centralized platform that provides customers an easy way to search and book vehicles online, provides branch staff with the information they require to approve bookings quickly and accurately, provides the fleet team with automated maintenance tracking and provides management with real-time financial data and functional reporting.

1.2 Objectives

- To provide consumers with a user-friendly online platform for searching, comparing and booking rental vehicles.
- To give branch staff real time visibility of vehicle availability, location and maintenance status across all branches.
- Automated the booking approval process by consolidating vehicle status, customer information and payment details on a single screen.
- Provide secure online payment processing, invoicing and refunds.
- To provide automated daily, monthly and yearly revenue and utilization reports to management.
- To reduce manual errors and double bookings created by disconnected branch-level recordkeeping.
- Maintain a complete vehicle service and maintenance history to support fleet management choices.

1.3 Target Users / Stakeholders

The system is meant for the following users: clients who rent vehicles, branch staff who approve bookings and handle handover/ return, fleet coordinators who handle vehicle maintenance, finance and admin managers who check revenue and create reports, operations managers who supervise multi-branch operations and system managers who manage accounts and systems configurations. Details are provided in section 5.

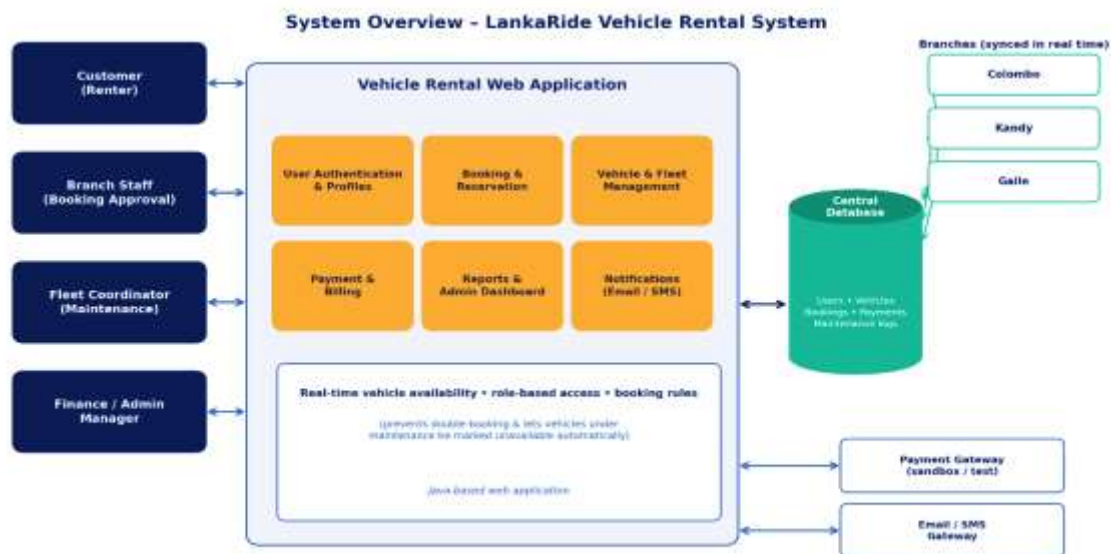
1.4 Scope and Limitations

This project is to develop and design a web application in java which includes features such as registration of users and authentication, automobile management, booking and reservation management, processing of payments, an administrative dashboard with reporting and customer support and alerts. During the semester the system will be built as an operational prototype based on an Agile (Scrum) approach.

The initial release of the system will concentrate on core rental operations for the three branches identified by the stakeholders, namely Colombo, Kandy and Galle. Native mobile applications, GPS based live vehicle monitoring and integration with third-party coverage providers are out of scope for this phase and are noted as potential future upgrades.

2. System Overview Diagram

The diagram below shows the high-level flow of the system: customers and branch employees interact with the Lanka Ride web app using a browser, the core modules of the application process the requests against a centralized database, the database keeps the vehicle and booking data synchronized between the Colombo, Kandy and Galle branches. The payment Processing and Notification modules interface with third-party services (a payment gateway and an email/SMS service) to finalize transactions and make sure the customer is notified.



3. Functional Requirements

3.1 Fleet Maintenance & Servicing Management

- Records vehicle mileage, fuel consumption and maintenance logs for all vehicles. Automatically sends maintenance reminders when time for services or based on mileage.
- Has a comprehensive maintenance record for each vehicle which records services, repairs, dates, cost, and any damage recorded.
- Staff note mileage and fuel level when collecting and returning the vehicle to check for any unusual fuel usage or damage.

- Automatically communicates with the Booking & Reservation module to ensure that vehicles under maintenance are unavailable and therefore not double booked.
- Fleet Coordinators can create review, update and close maintenance/service records (CRUD) and access the full-service record for any vehicle anytime.

3.2 Vehicle Management

- Fleet coordinators and administrators can create, view, update and remove vehicle information (category, brand, model, seats, gearbox, fuel type, features, photos and price/day, deposit amount, location).
- For every vehicle transferred between branches, System records it, guaranteeing employees to know where a vehicle is. If a vehicle is moving, damaged or undergoing maintenance, it will automatically be placed on the unavailability list.

3.3 Booking & Reservation Management

- Vehicles can be searched and filtered by type, price, gearbox, fuel type, branch and data range.
- Before making a reservation, customers may check all the vehicle's details, including images, pricing per day, seats, deposit and extras like GPS or air conditioning.
- Clients can make reservations for specific dates and pickup location.
- All information required to approve a booking, including vehicle availability, customer information and rental history, vehicle status and deposit/advance payment status, may be viewed on a single screen by branch employees.
- A booking request may be accepted or denied by branch employees.
- The booking status (Pending, Approved, ongoing, completed, cancelled) is monitored by the platform.

3.4 Payment & Billing Management

- Booking deposits and rental fees can be safely paid online by customers.
- For transactions that are finished, the system creates invoices.
- Refunds for authorized cancellations are processed by the system.
- Clients have access to their transaction and payment history.
- When a car is returned, the system determines any applicable late fees or damage charges.

3.5 Admin Dashboard & Reporting

- Daily reports (rentals, transactions received, deposits collected, refunded payments, income) are generated by the system.
- Monthly statements (total revenue, booking, outstanding payments, refunds and most-rented categories) are produced by the system.
- Annual reports (overall performance, profit/revenue trends, branch-based comparison, vehicle utilization) are generated by the system.
- Instead of creating reports by hand, managers can view important operational and financial data on a dashboard.

3.6 Customer Support & Notifications

- Booking approvals, payment receipts and reminders are automatically sent by the system via SMS and email.
- Consumers can submit support enquiries, which employees can review and address.

4. Non – Functional Requirements

Quality Attribute	Description
Performance	Under typical traffic, core sites (search, booking, dashboard) should load two to three seconds availability updates should appear almost instantly across branches.
Security	Role based access control is implemented, passwords are secured, and payment information is managed securely.
Usability	Customers and branch employees should find the interface easy to use and consistent, requiring few steps to finish a booking or approval.
Reliability	By using uniform, synchronized availability data across branches, the system should stop double booking.
Availability	The booking gateway that interacts with customers should be available around the clock.
Scalability	New branches, cars, and users should be able to be added to the system without requiring a significant overhaul.
Maintainability	Future modifications should be made easier with a modular design that includes distinct booking, fleet, payment, and reporting modules.
Compatibility	The system should be responsive on desktop and mobile displays and compatible with the majority of platforms.

5. Major Stakeholders / Users

Structured interviews with Lanka Ride rentals employees, including the Operations Manager, Branch Supervisor, Customer Service Officer, Finance Manager, and Vehicle fleet coordination, generated the stakeholder needs listed below.

Role	Responsibilities	Key Needs
Customer (Renter)	Books, Browses, and makes car rental payments	Easy booking, prompt alerts, and transparent pricing and availability
Branch Staff (Booking Supervisor)	Verifies vehicle and customer information, authorizes reservation, and oversees handover and return	Quick decision making with all approved data on one screen
Fleet Coordinator	Oversees the scheduling of maintenances, condition monitoring, and vehicle inventory	Complete service history, mileage and fuel information, and automatic maintenance reminders
Finance/Admin Manager	Oversees financial reporting, revenue, and expenditures	A real time dashboard with daily, monthly, and annual data
Operations Manager	Oversees the distribution of vehicles and multi branch operations	Availability and location visibility in real time across locations
System Administrator	Oversees system setup, roles and user profiles	System dependability and secure control of access

6. Six Major Functions

Following the modules requirement that each member deliver their own functioning during the proposal presentation, each of the six key functions listed below is mapped one to one to a group member. This is recommended division based on how the functions were determined during requirement gathering before submitting, switch members names as the organization sees fit.

Member	Major Function	Who uses it	What it does	Expected outcome	CRUD operation
DE SILVA D.L.K.C	Fleet Maintenance & Servicing Management	Fleet Coordinator	The features relevant to their job are only available for verified and authorized users.	Keeps vehicles safe and in good working order. Automatically notifies the Booking system that vehicles in Maintenance are not available for booking, thus avoiding double bookings.	Create: Record new maintenance/service records. Read: Check out service history and next service schedule. Update: Update vehicle maintenance status and availability. DELETE: Remove old or redundant information.
SAKALASOORIYA S.A.A. A	Vehicle Management	Operation managers and fleet coordinators	Create, update and remove vehicles; know which branch is available and locate in real time throughout the fleet.	The present spreadsheet-based uncertainty is eliminated because staff members are always aware of which cars are available, where they are and their condition	Create: Add Vehicle Records. Read: Check fleet listing and availability. Update: Edit vehicle information, location and status. Remove: take off retired vehicles
SAMARANAYAKE P.I. S	Reservation and Booking Management	Clients and branch employees	All the booking-related details	Staff can swiftly authorize	Create: Make new bookings/Reservations. Read – See the vehicle,

			(vehicle, customer, etc. payment status) are visible in one screen.	reservations without double booking a car, and customers may make confident reservations	customer and booking information. Update: Approve or modify/change the status of the booking. Delete: Cancel or remove expired bookings.
KAVINDI P.D.N	Management s of payments and billing	Clients and financial managers	Safe online transaction handling, creation of invoices, refunds, payments of deposits and fees.	With a visible audit trail, rental payments are managed consistently and transparently	Create: Make invoices and payment records. Read: check payments and transactions. Update: Update Refund/Deposit Status. Delete: remove failed or duplicate transactions.
WICKRAMASINGHE R.D.W.K.G. S	Reporting and admin dashboard	Operation managers and finance/admin managers	In real-time, displays of vital operational and financial information.	Without having to manually compile reports, management may make well informed judgements	Create: produce daily, monthly and annual reports. Read: Display the metrics and trends on the dashboard. Update: Change report parameters/filter. Remove: Get rid of old reports. Reports from daily, monthly and annually can be used to make informed decision for management.
PAHASARA Y.A.P	Customer service and alerts	Clients and customer service representatives	Email or SMS booking and payment notifications —also used to submit and track support enquiries, automated.	Consumers are kept informed throughout the entire renting process, and support enquiries are managed centrally	Create: create support tickets and notifications. Read – Enquiry status and history view. Update: Update answers to tickets. Delete: Close or delete resolved tickets.

7. Minor Functions

Instead of being owned by a single main function, these supporting roles are utilized across the system:

- Logging in and out.
- Account recovery and password reset.
- Managing customer profiles, including checking rental history and amending personal information.
- Vehicle search and filter capabilities (by type, price and availability).
- Automated maintenance alerts based on service dates and vehicle miles.
- System-wide alerts for reminders, payment receipts and booking confirmations.
- Role-based access control for administrators, employees and customers.

8. System Limitations / Constraints

- Native mobile apps are outside the purview of this phase; the system will be created as a Java-based web application that can be accessed using common web browsers.
- To access booking and payment capabilities, the system requires users to have a reliable internet connection.
- The three branches – Colombo, Kandy and Galle – identified during stakeholder interviews will be supported by the initial version.
- The availability and dependability of a third-party payment platform will determine how online payments are processed.
- The scope of features that can be fully created and tested is restricted by the project's one-semester timeline.
- The system depends on people to maintain correct and current vehicle and maintenance data; delays in human updates may still have an impact on real-time correctness.

9. Project Timeline

Week	Planned Activities
Week 3	Present the proposal, submit the report, and complete the requirements.
Week 4-5	UI wireframes, database scheme design, and use case and class diagrams.
Week 6-7	Implement authentication and vehicle/ fleet management modules in the first sprint.
Week 8-9	Implement the payment/ billing and booking/ reservation modules in sprint 2.
Week 10	Presentation of progress and submission of design documents (about 75% finished).
Week 11	Implement the admin dashboard, reporting and notification modules in sprint 3.
Week 12	Implementation of design patterns, problem fixes, and integration testing.
Week 13	Final presentation and viva, refine the demo flow and user interface.
Week 14	Project completion and submission of the final report.

10. Conclusion

The Web-Based Vehicle Rental System solves actual, verifiable issues at Lanka Ride Rentals, including time-consuming manual reporting, limited visibility of vehicle location and maintenance status and duplicate reservations brought on by unconnected branches. The system will enable customers to make confident reservations, staff to approve reservations from a single screen, the fleet team to stay ahead of maintenance and management to view performance in real time by centralizing vehicle, booking, payment and maintenance data across the Colombo, Kandy and Galle locations.

The project is possible within the module timetable because the phased Agile/ Scrum plan (Section 6 and 9) enables the team to develop, integrate and show incremental progress at each assessment checkpoint and the key modules neatly map into the major functions of the six group members.